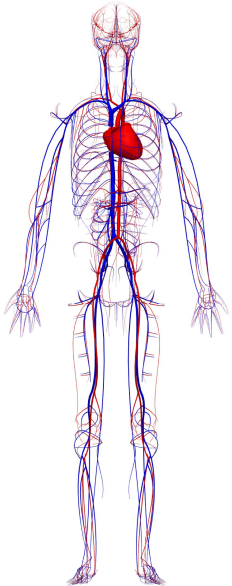


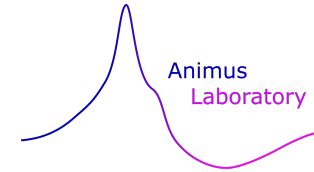
Circulatory Autogen: a software for coupling CellML modules and calibrating to data



Presented By
Finbar Argus and Beatrice Ghitti

Member of
Animus Lab
The Auckland Bioengineering Institute
Oxford DPAG (former)
Oxford IDRM (former)

Funded By
The Andrew Bagnall Fellowship
12 Labours
VITAL Project
MBIE Soul Machines



Circulatory Autogen Aims

The 12 Labours Project

VITAL Horizon Europe Project

Aims

1. Allow for easy coupling of CellML modules and calibration of the generated models to anatomical and functional clinical data.
2. Allow calibration of general CellML models to experimental data.



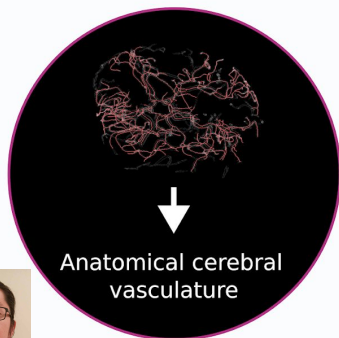
Aim 1

Allow for easy coupling of CellML modules and calibration of the generated models to anatomical and functional clinical data.

- Vessel names
- Vessel types
- Vessel connections
- Boundary conditions
- Parameters

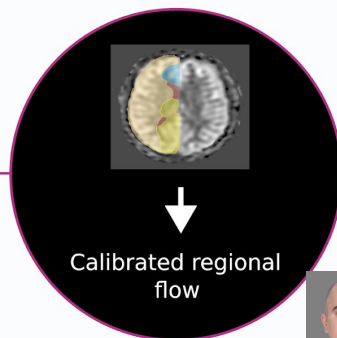
- Dynamic data
- Anatomical Images
- Patient metrics

Bayesian
Uncertainty
Quantification



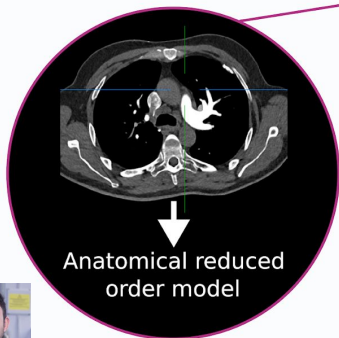
Anatomical cerebral
vasculature

Cerebral MRA



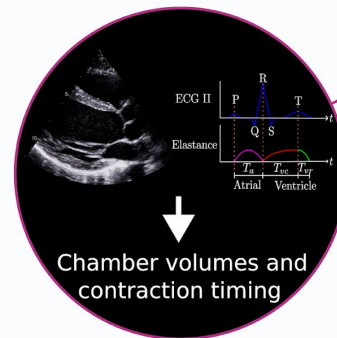
Calibrated regional
flow

Cerebral ASL



Anatomical reduced
order model

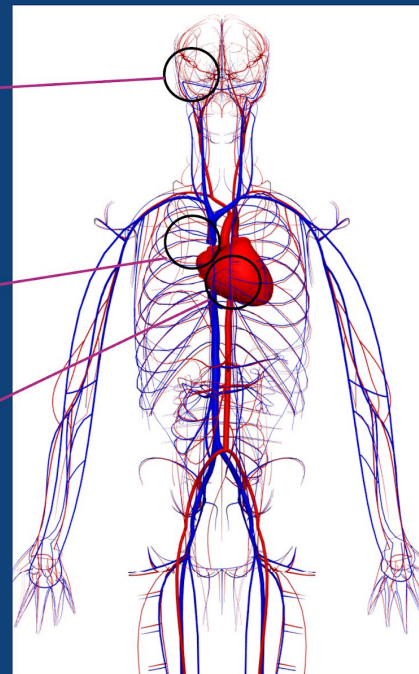
Pulmonary CTPA



Chamber volumes and
contraction timing

Echo + ECG

Calibrated CVS Model



Software Structure



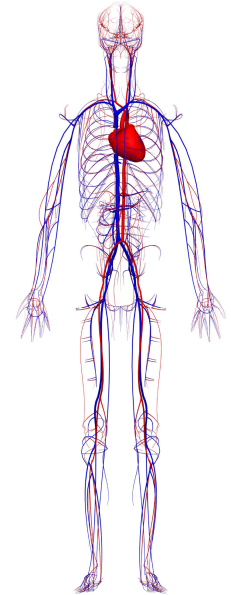
`model_vessel_array.csv`
+
`model_parameters.csv`

Module
configuration files

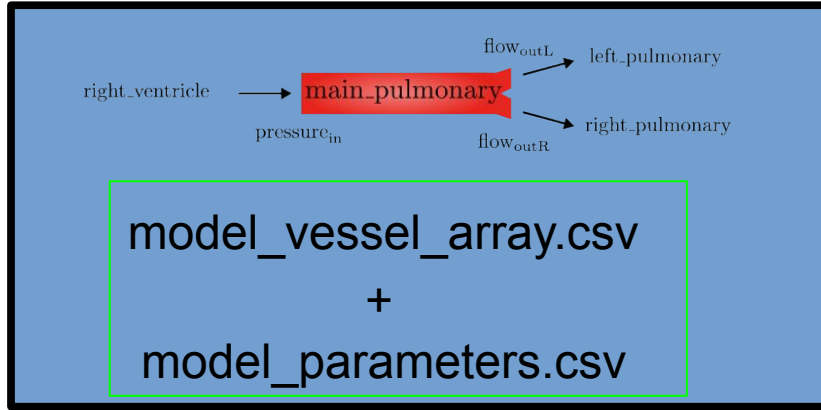


CellML modules
LibCellML

Generated Model



Software Structure

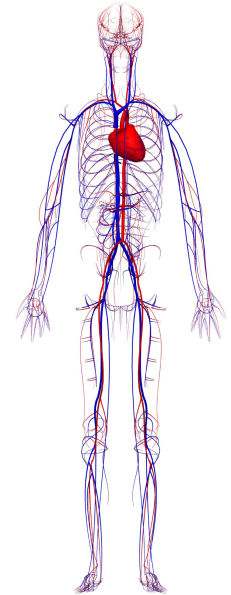


Module
configuration files



CellML modules
LibCellML

Generated Model



Software Structure



model_vessel_array.csv

name	BC_type	vessel_type	inp_modules	out_modules
main_pulmonary	pv	arterial	right_ventricle	left_pulmonary right_pulmonary
...	...	...	...	...

model_parameters.csv

Software Structure



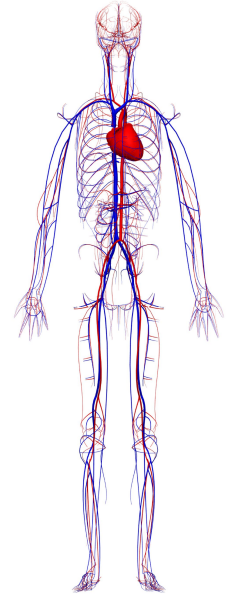
model_vessel_array.csv
+
model_parameters.csv

Module
configuration files



CellML modules
LibCellML

Generated Model



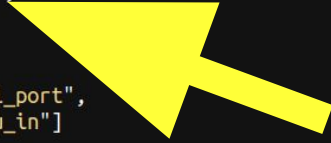
Module configuration files

```
{
  "vessel_type": "arterial",
  "BC_type": "pv",
  "module_format": "cellml",
  "module_file": "BG_modules.cellml",
  "module_type": "pv_type",
  "entrance_ports":
  [
    {
      "port_type": "vessel_port",
      "variables": ["v", "u_in"]
    }
  ],
  "exit_ports":
  [
    {
      "port_type": "vessel_port",
      "variables": ["v_out", "u"]
    },
    {
      "port_type": "flow_port",
      "variables": ["v"]
    }
  ]
}
```

```
,
  "variables_and_units":
  [
    ["u", "J_per_m3", "access", "variable"],
    ["v", "m3_per_s", "access", "variable"],
    ["E", "J_per_m3", "access", "constant"],
    ["l", "metre", "no_access", "constant"],
    ["r", "metre", "access", "constant"],
    ["u_ext", "J_per_m3", "access", "constant"],
    ["theta", "dimensionless", "no_access", "constant"],
    ["beta_g", "dimensionless", "no_access", "global_constant"],
    ["rho", "Js2_per_m5", "no_access", "global_constant"],
    ["mu", "Js_per_m3", "no_access", "global_constant"],
    ["g", "m_per_s2", "no_access", "global_constant"],
    ["a_vessel", "dimensionless", "no_access", "global_constant"],
    ["b_vessel", "per_m", "no_access", "global_constant"],
    ["c_vessel", "dimensionless", "no_access", "global_constant"],
    ["d_vessel", "per_m", "no_access", "global_constant"],
    ["u_in", "J_per_m3", "access", "boundary_condition"],
    ["v_out", "m3_per_s", "access", "boundary_condition"]
  ]
},
```

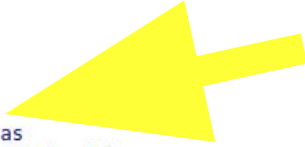
Module configuration files

```
{
  "vessel_type": "arterial",
  "BC_type": "pv",
  "module_format": "cellml",
  "module_file": "BG_modules.cellml",
  "module_type": "pv_type",
  "entrance_ports":
  [
    {
      "port_type": "vessel_port",
      "variables": ["v", "u_in"]
    }
  ],
  "exit_ports":
  [
    {
      "port_type": "vessel_port",
      "variables": ["v_out", "u"]
    },
    {
      "port_type": "flow_port",
      "variables": ["v"]
    }
  ]
}
```



```
},
"variables_and_units":
[
  ["u", "J_per_m3", "access", "variable"],
  ["v", "m3_per_s", "access", "variable"],
  ["E", "J_per_m3", "access", "constant"],
  ["l", "metre", "no_access", "constant"],
  ["r", "metre", "access", "constant"],
  ["u_ext", "J_per_m3", "access", "constant"],
  ["theta", "dimensionless", "no_access", "constant"],
  ["beta_g", "dimensionless", "no_access", "global_constant"],
  ["rho", "Js2_per_m5", "no_access", "global_constant"],
  ["mu", "Js_per_m3", "no_access", "global_constant"],
  ["g", "m_per_s2", "no_access", "global_constant"],
  ["a_vessel", "dimensionless", "no_access", "global_constant"],
  ["b_vessel", "per_m", "no_access", "global_constant"],
  ["c_vessel", "dimensionless", "no_access", "global_constant"],
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  ["u_in", "J_per_m3", "access", "boundary_condition"],
  ["v_out", "m3_per_s", "access", "boundary_condition"]
]
```

Connection to a CellML module



```
def comp pv_type as
  var t: second {pub: in};

  // Parameters

  var mu: Js_per_m3 {pub: in};
  var rho: Js2_per_m5 {pub: in};
  var g: m_per_s2 {pub: in};
  var beta_g: dimensionless {pub: in};
  var theta: dimensionless {pub: in};
  var E: J_per_m3 {pub: in};
  var l: metre {pub: in};
  var h: metre {pub: out};
  var r: metre {pub: in};
  var I: Js2_per_m6 {pub: out};
  var C: m6_per_J {pub: out};
  var R: Js_per_m6 {pub: out};
  var R_v: Js_per_m6 {pub: out};
  var a_vessel: dimensionless {pub: in};
  var b_vessel: per_m {pub: in};
  var c_vessel: dimensionless {pub: in};
  var d_vessel: per_m {pub: in};
```

```
// Variables

var u_ext: J_per_m3 {pub: in};
var u_in: J_per_m3 {pub: in};
var v: m3_per_s {init: 0.0, pub: out};
var u: J_per_m3 {pub: out};
var u_C: J_per_m3 {pub: out};
var v_out: m3_per_s {pub: in};
var q_C: m3 {init: 0.0, pub: out};

// Phenomenological Relations

h = r*(a_vessel*exp(b_vessel*r)+c_vessel*exp(d_vessel*r));
I = rho*/(pi*sqr(r));
C = 2{dimensionless}*pi*pow(r, 3{dimensionless})*l/(E*h);
R = 8{dimensionless}*mu*/(pi*pow(r, 4{dimensionless}));
R_v = 0.01{second}/C;

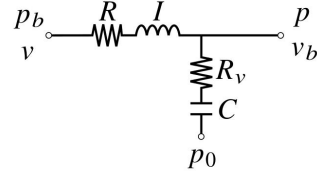
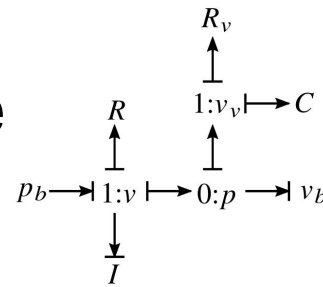
// Conservation Laws

ode(v, t) = (u_in-u-R*v-beta_g*rho*g*l*cos(theta*pi/180{dimensionless}))/I;
ode(q_C, t) = v-v_out;

// Constitutive equations

u_C = q_C/C+u_ext;
u = u_C+R_v*(v-v_out);
```

```
enddef;
```



Software Structure



`model_vessel_array.csv`
+
`model_parameters.csv`

Generated Model

+

`ground_truth_data.json`

+

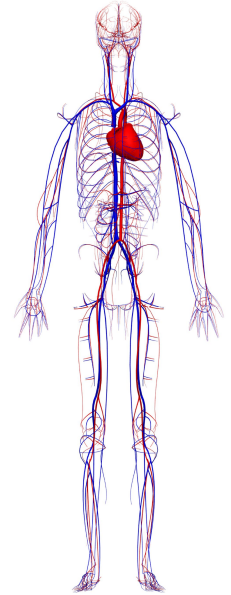
`params_for_id.csv`

Module
configuration files



CellML modules
LibCellML

Generated Model



Bayesian Inference
Methods



Calibrated Model

params_for_id.csv

```
1 vessel_name,      param_name,      min,      max,      name_for_plotting
2 heart,           q_lv,           14e-4,    40e-4,    q_{sbv}
3 leg_L_T leg_R_T,   R_T,       6e5,      5e9,      R_{Tleg}
4 arm_L_T arm_R_T,   R_T,       6e5,      5e10,     R_{Tarm}
5 head_L_T head_R_T, R_T,       6e5,      5e10,     R_{Thead}
6 trunk_C_T,       R_T,       6e5,      5e9,      R_{Ttrunk}
```

params_for_id.csv

1	vessel_name,	param_name,	min,	max,	name_for_plotting
2	heart,	q_lv,	14e-4,	40e-4,	q_{sbv}
3	leg_L_T leg_R_T,	R_T,	6e5,	5e9,	R_{Tleg}
4	arm_L_T arm_R_T,	R_T,	6e5,	5e10,	R_{Tarm}
5	head_L_T head_R_T,	R_T,	6e5,	5e10,	R_{Thead}
6	trunk_C_T,	R_T,	6e5,	5e9,	R_{Ttrunk}

ground_truth_obs_data.json

```

14  { "variable": "flow aortic root",
15    "name_for_plotting": "v_{AR}",
16    "data_type": "constant",
17    "operation": "max",
18    "operands": ["aortic_root/v"],
19    "unit": "m3_per_s",
20    "weight": 1.0,
21    "value": 5e-4,
22    "std": 5e-5,
23    "cost_type": "gaussian_MLE",
24    "plot_type": "horizontal"
25  },
26  { "variable": "stroke volume",
27    "name_for_plotting": "q_{lv}",
28    "data_type": "constant",
29    "operation": "max_minus_min",
30    "operands": ["heart/q_lv"],
31    "unit": "m3",
32    "weight": 2.0,
33    "value": 1.04e-4,
34    "std": 5e-5,
35    "cost_type": "gaussian_MLE",
36    "plot_type": "horizontal_from_min"

```

Software Structure



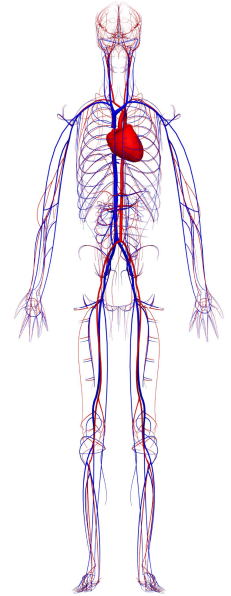
model_vessel_array.csv
+
model_parameters.csv

Generated Model
+
ground_truth_data.json
+
parameters_to_identify.csv

Module
configuration files

CellML modules
LibCellML

Generated Model



Optimisation Algorithms

[Lib]OpenCOR

Calibrated Model

Software Structure



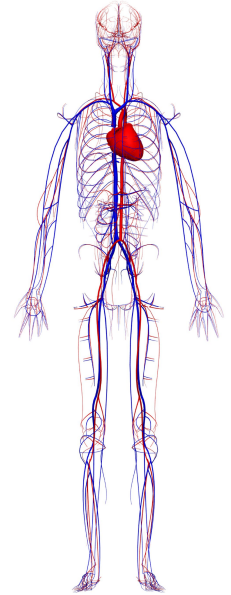
model_vessel_array.csv
+
model_parameters.csv

Generated Model
+
ground_truth_data.json
+
parameters_to_identify.csv

Module
configuration files

CellML modules
LibCellML

Generated Model

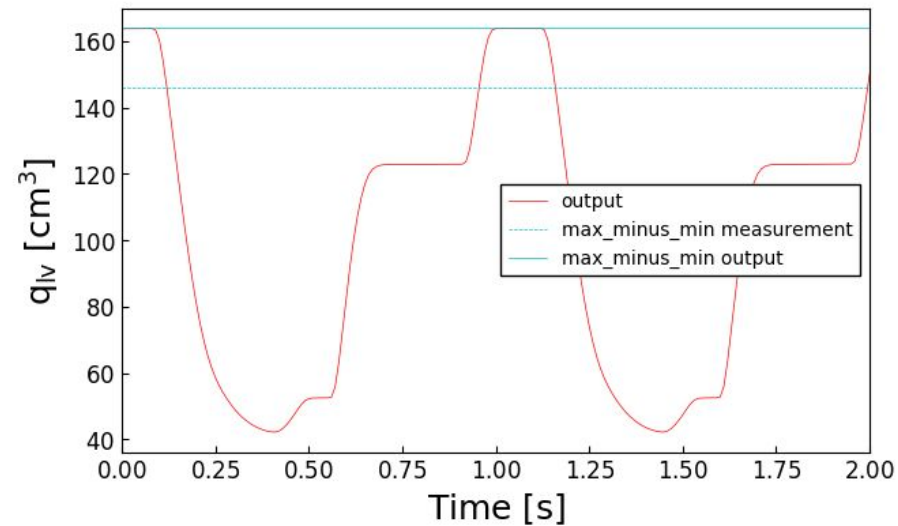
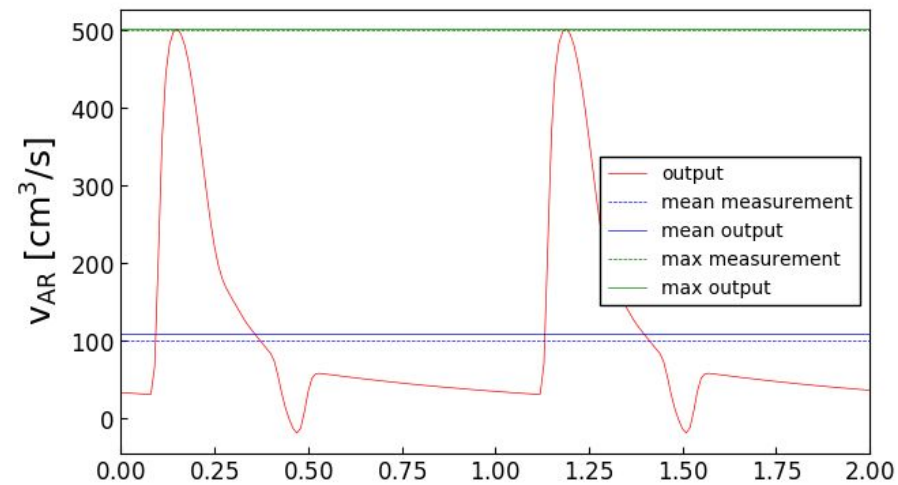
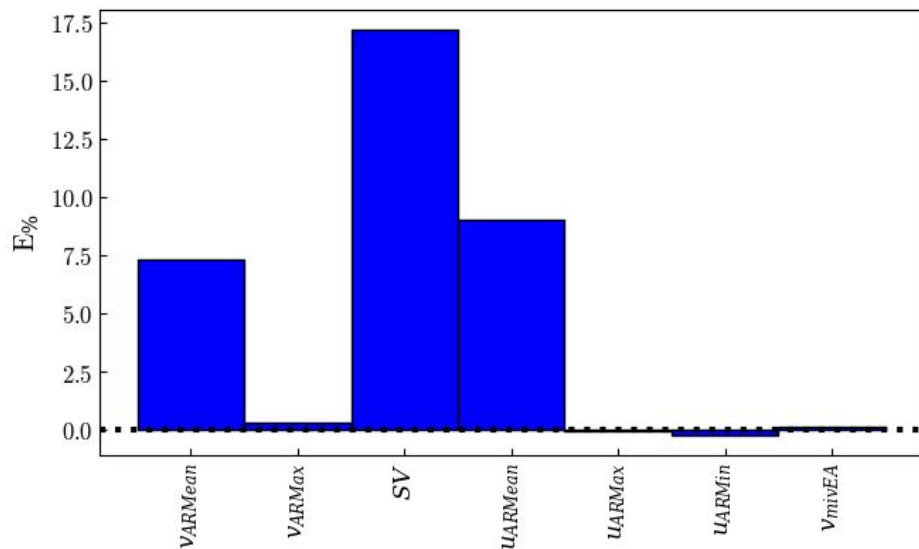


Optimisation Algorithms

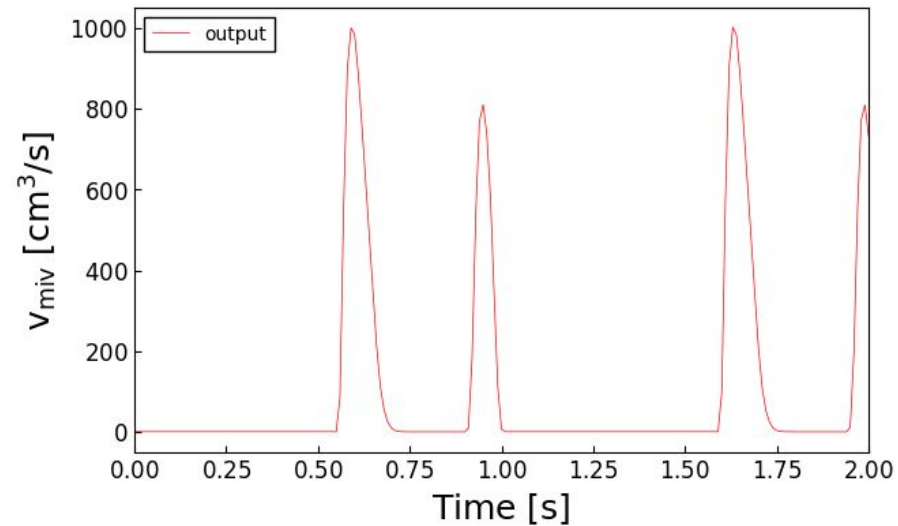
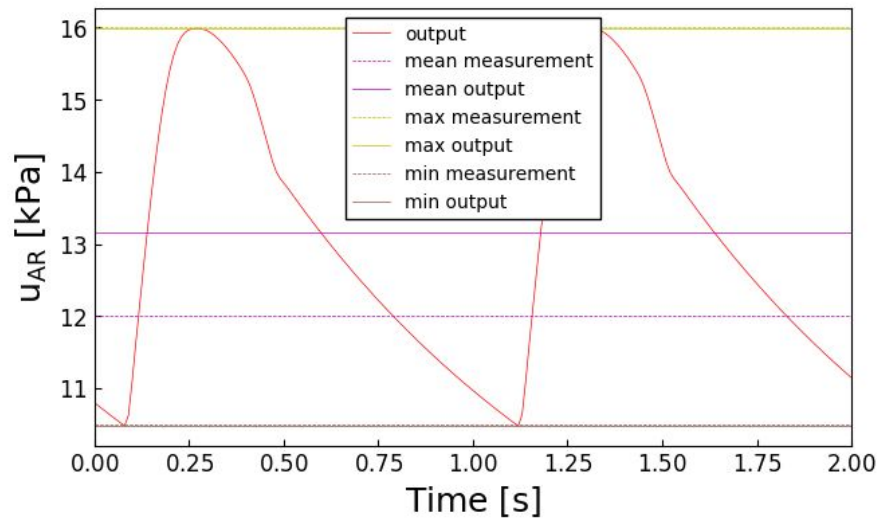
[Lib]OpenCOR

Calibrated Model

Calibrated Model



Calibrated Model

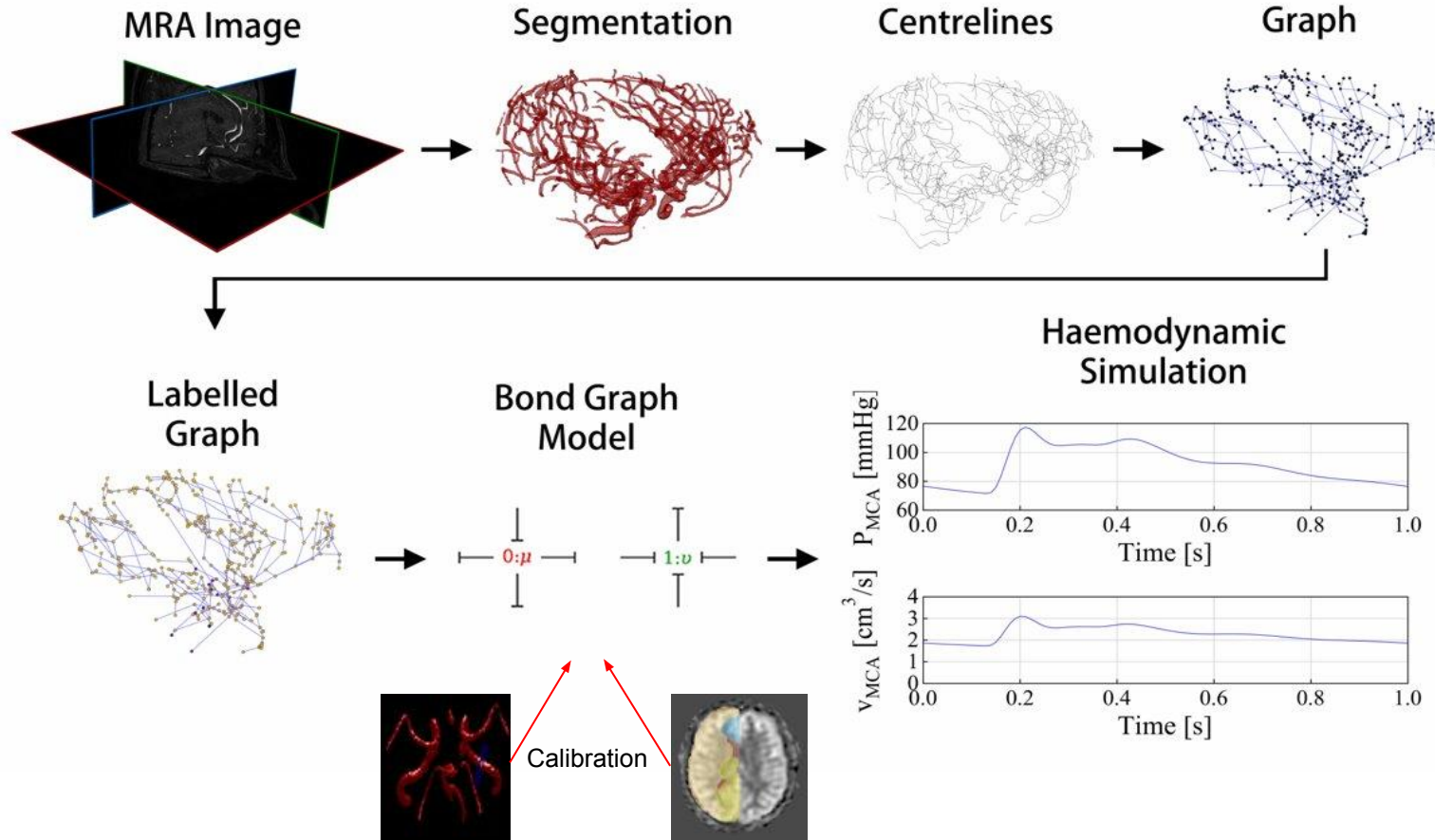


Example Uses

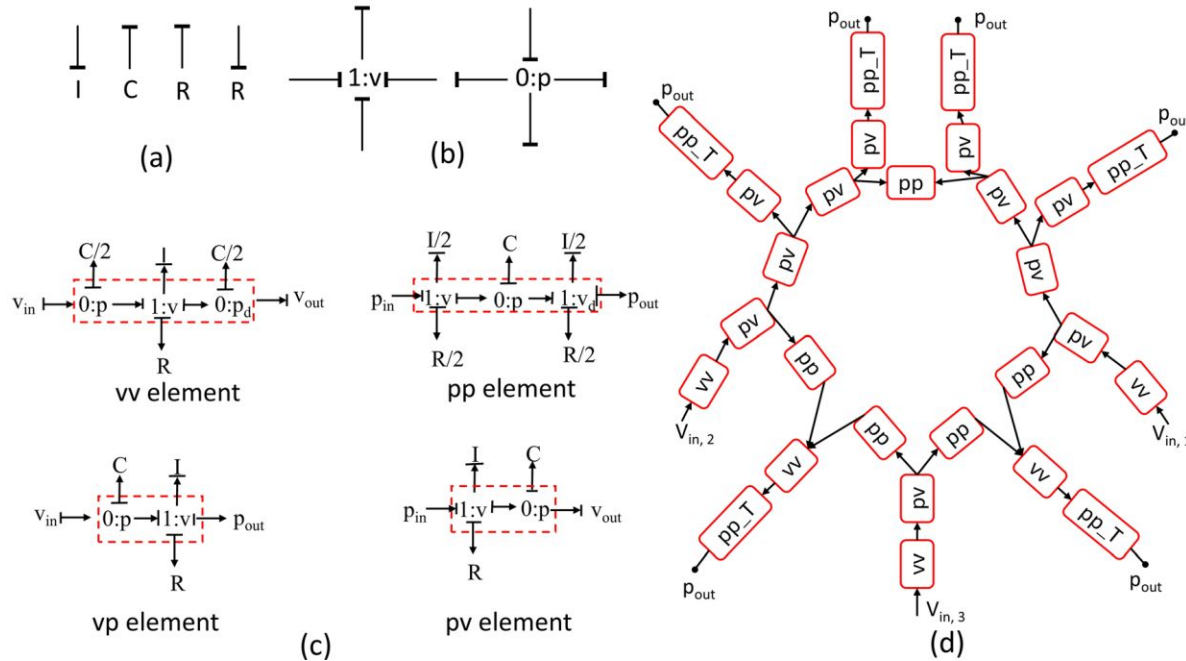
Case 1: MR to CVS predictions



Animus
Laboratory

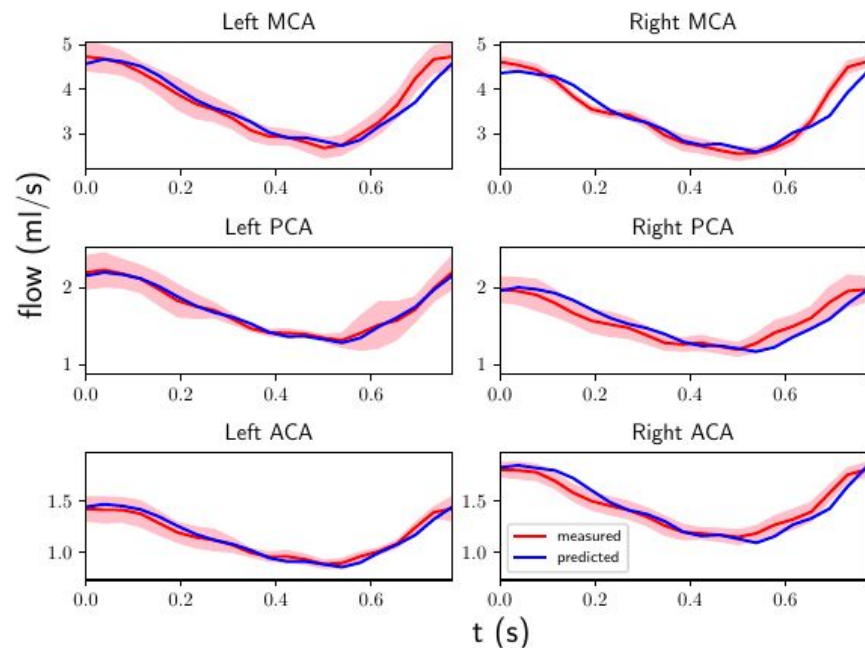


Case 1: MR to CVS predictions



Case 1: MR to CVS predictions

Personalised Calibration



Personalised Prediction

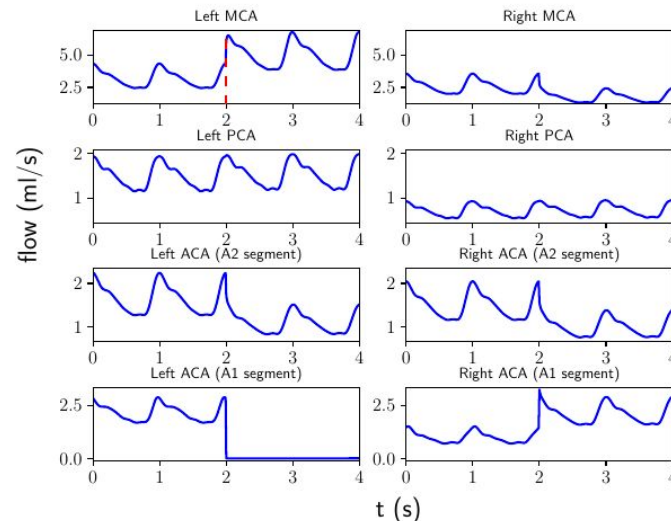
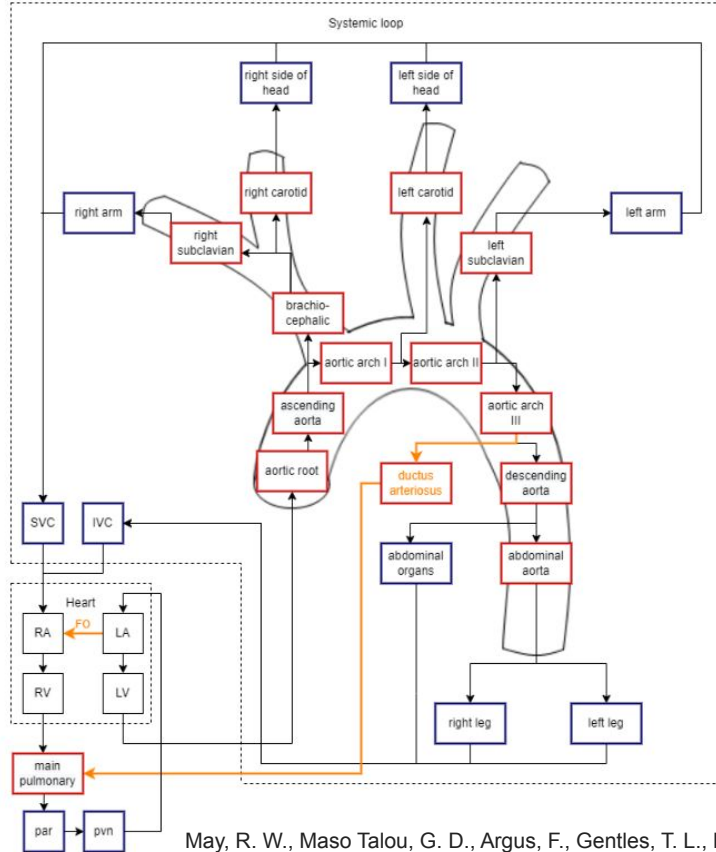
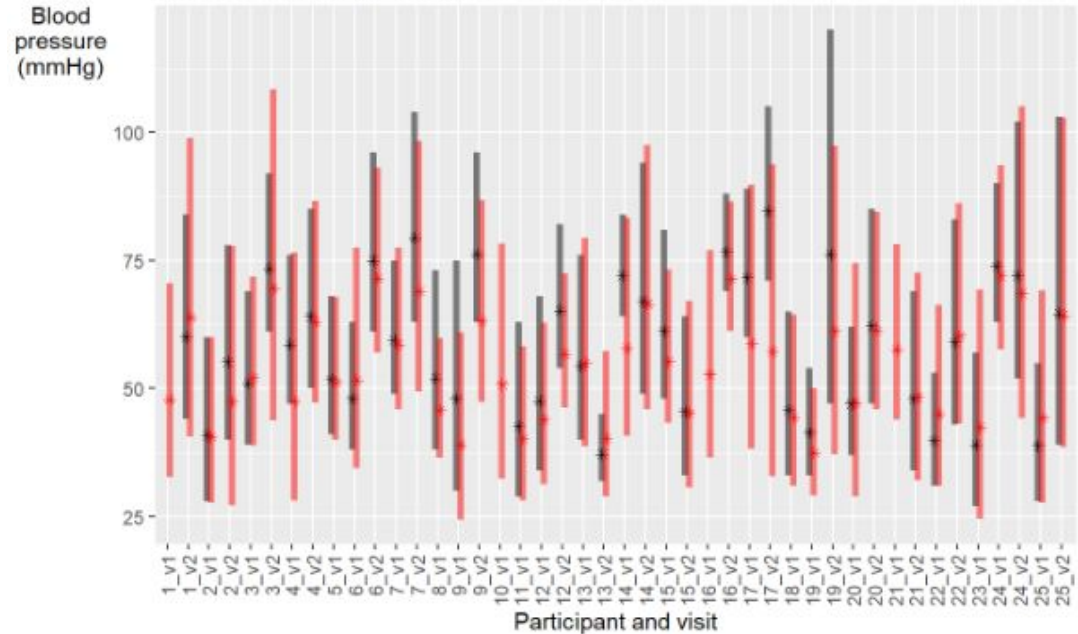


Fig. 8: Predicted flow of Subject 2 before and after being subjected to A1 segment blockage occurring at time = 2s. This case study predicts that the blockage will cause Subject 2 to have an overperfused Left MCA and underperfused Right MCA, Left ACA, and Right ACA

Case 2: Newborn CVS modelling



Measured blood pressures vs prediction 1



May, R. W., Maso Talou, G. D., Argus, F., Gentles, T. L., Bloomfield, F. H., & Safaei, S. (2023, June). An Image-Based Computational Model of the Newborn Cardiovascular System with Term and Preterm Applications. In *International Conference on Functional Imaging and Modeling of the Heart* (pp. 475-484). Cham: Springer Nature Switzerland.

Case 3: Sympathetic Neuron Model

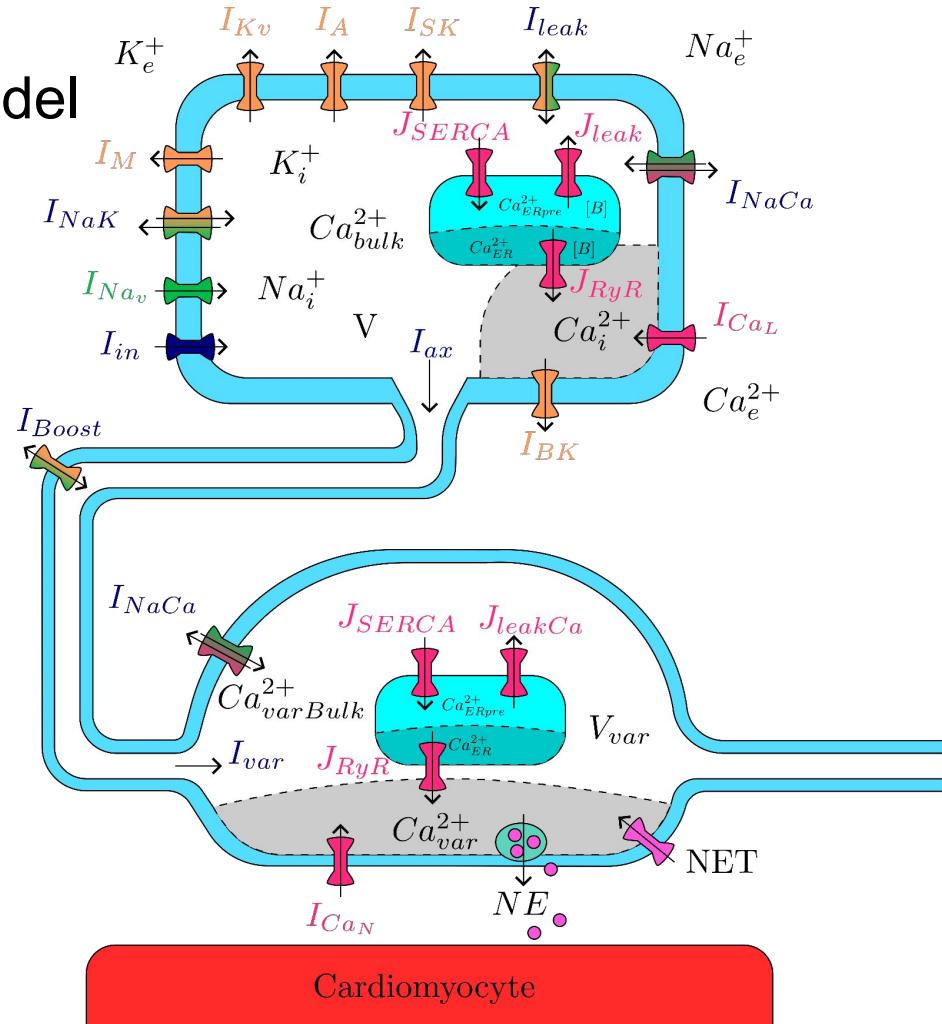
Vessel Array File

	A	B	C	D	E
1	name	BC_type	vessel_type	inp_vessels	out_vessels
2	soma_SN	nn	SN_soma		axon_SN
3	axon_SN	nn	SN_axon	soma_SN	var_SN
4	var_SN	nn	SN_varicosity	axon_SN	

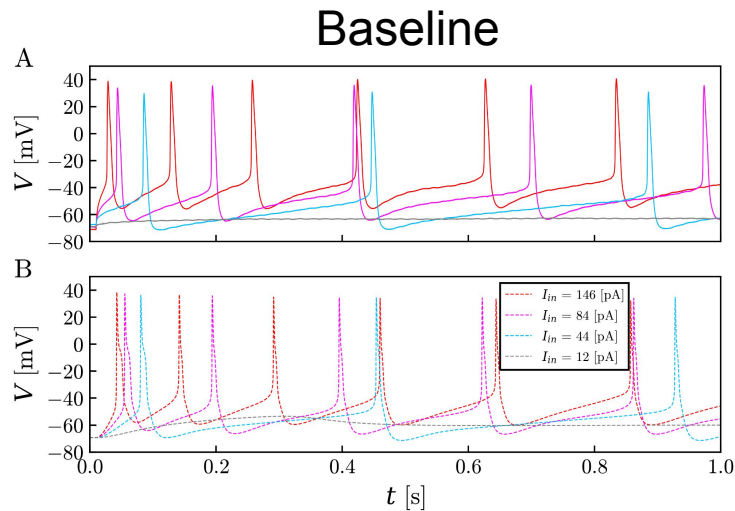
Parameters File

	A	B	C	D
1	variable_name	units	value	data_reference
2	g_Na_soma_SN	microS	2.3	user_defined
3	g_Na_mod_soma_SN	dimensionless	1	user_defined
4	g_Na_soma_SN_OLD	microS	9.28	Tau2011
5	Cm_soma_SN	picoF	30	Davis2021Transcriptomic
6	Cm_SN_OLD	picoF	240	Tau2011
7	p_Ca_soma_SN	m3_per_s	4E-14	to_be_identified
8	g_Kv_soma_SN	microS	0.0013333	Davis2021Thesis_fit_to_curve
9	g_A_soma_SN	microS	0.08878493886	to_be_identified
10	g_KCa_BK_soma_SN	microS	0.2	was70microS_per_cm2_inKhaliq2003
11	g_KCa_SK_soma_SN	microS	0.005	to_be_identified
12	g_KCa_SK_mod_soma_SN	dimensionless	1	to_be_identified
13	Vol_soma_SN	m3	2E-15	Wu2012Soma_Approx
14	Cai_init_soma_SN	millimolar	0.0001	Hagiwara1981_was_1e-4_change_to_allow_buffering_decrease
15	Na_i_init_soma_SN	millimolar	13	BoronBoulpaep2017_was_15
16	Ki_init_soma_SN	millimolar	150	BoronBoulpaep2017
17	Cao_soma_SN	millimolar	2	Tau2011
18	Nao_soma_SN	millimolar	145	BoronBoulpaep2017
19	Ko_soma_SN	millimolar	4.5	BoronBoulpaep2017

...

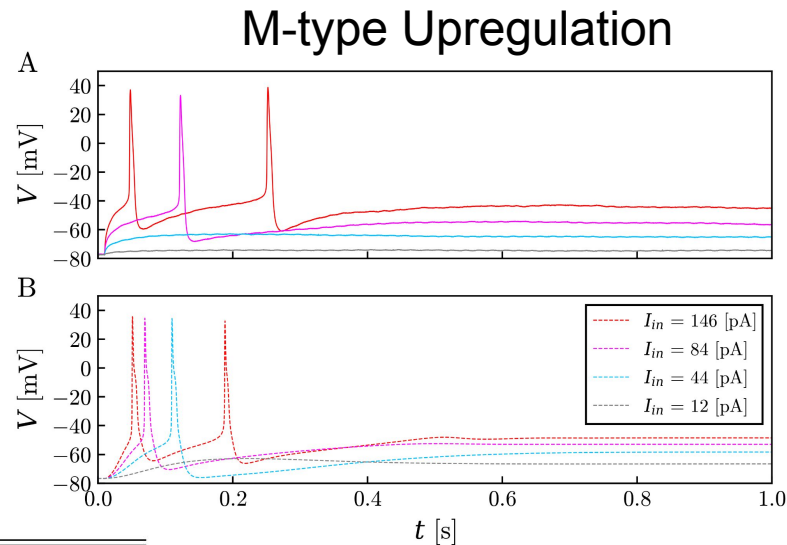


Case 3: Sympathetic Neuron Model: Calibration



Experimental
Results

Simulation
Results



	V_{max} %	V_{min} %	f %	T_{min} %
Baseline 12pA	-14.3	-5.4	0.0	0.0
Baseline 44pA	17.7	1.2	0.0	3.3
Baseline 84pA	3.4	4.9	0.0	-7.9
Baseline 146pA	-5.4	8.9	0.0	-1.2
M-type 12pA	-14.8	-11.3	0.0	0.0
M-type 44pA	-154.4	-6.1	100.0	0.0
M-type 84pA	3.8	-7.0	0.0	0.0
M-type 146pA	-8.1	8.5	0.0	-32.8

Conclusions

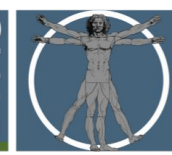
- We have developed Circulatory_Autogen for generating and calibrating system models including:
 - Cardiovascular System Models
 - Cell models (general CellML models)
 - Calibration to functional clinical data
 - Calibration to structural clinical data
 - Calibration to experimental data
- The software allows the following user-friendly features:
 - User-defined operations in python
 - User-defined cost-functions in python
 - Port boundary variables to ensure energy conserving coupling, based on bond-graph.
 - MCMC for uncertainty quantification of parameters and predictions.
- **We encourage use of these open-source tools by other groups and projects.**



Next Steps and Discussion Points

- **Ensure modules in the Physiome Repository have their port variables defined.**
- Automatic gradients through CVODES for efficient model calibration.
- Coupling to 1D blood flow models.
- More advanced priors for parameters to identify.
- Switch to LibOpenCOR: User defined external variables, random number generator, data inputs, etc.
- User defined optimisation algorithms (popular Python libraries, Hamiltonian Monte Carlo).
- Link with FC maps for automatic model generation from the web application.
- Name change to PhysiomeGen





Acknowledgements

12 Labours EP1 Team

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Alys Clark
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12 Labours P1 Team

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Jagir Hussan

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David Brooks

VITAL

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Chenchen Zhang
Jenny Wang
Blanca Rodriguez

UCL

Harvey Davis

The Andrew Bagnall Fellowship

The 12 Labours Project

SPARC Project

VITAL Horizon Europe Project

MBIE Soul Machines

BHF

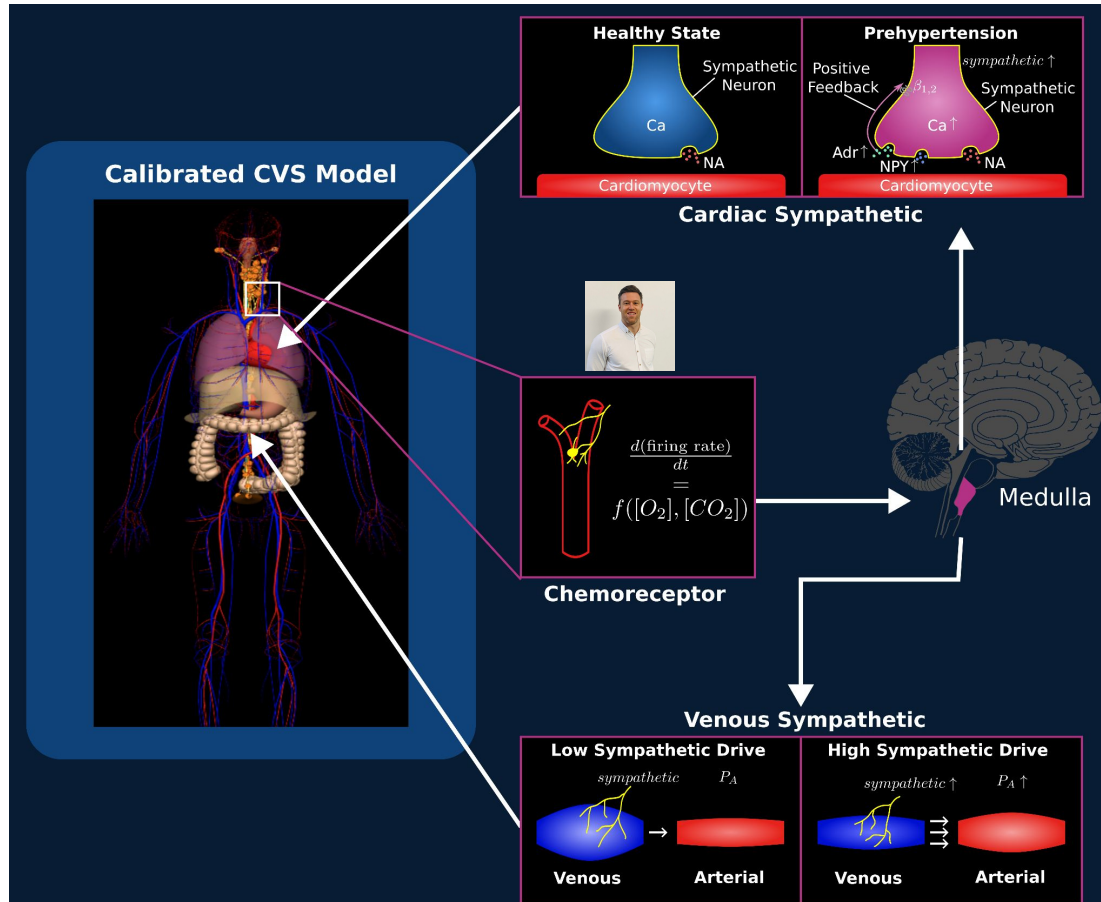
Leducq Foundation



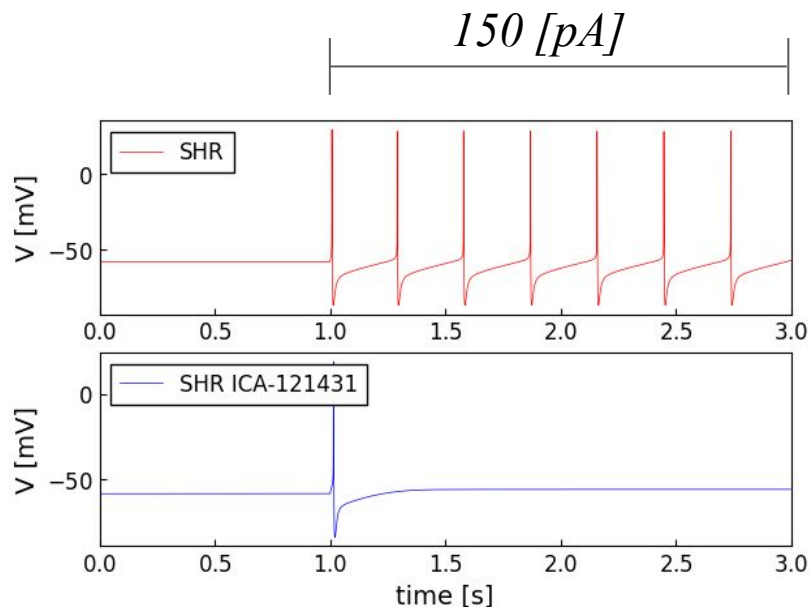
Software link: https://github.com/FinbarArgus/circulatory_autogen/tree/devel

Presenter Email: finbar.argus@auckland.ac.nz

Physiological ANS models



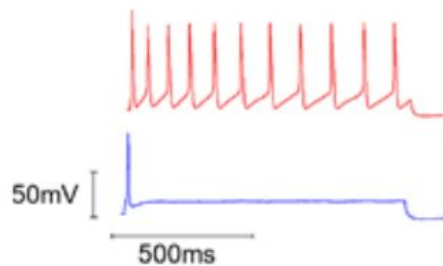
Experimental Calibration Na channel block



*ICA-121431 simulated by decreasing Na channel permeability

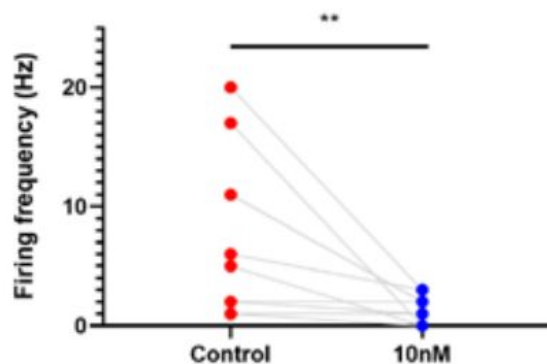
Simulation

150 [pA]



SHR

SHR +
ICA-121431



Downregulation of M Current Is Coupled to Membrane Excitability in Sympathetic Neurons Before the Onset of Hypertension